NOTICE

AI Half-Yearly Syllabus — Class XI (2026)

This is to inform all students that the Half-Yearly Examination for Artificial Intelligence will be conducted soon. Students are advised to prepare thoroughly from the topics listed below, which include the Short Test syllabus in addition to the new units.

1 Basic Concepts of AI

Introduction	‣ Definition, evolution, and applications of AI ‣ Healthcare, Data Analysis, and Problem Solving use-cases
Data & Computing	‣ Types of data ‣ Binary logic and conditional gates ‣ Deterministic vs. probabilistic problems
Decision Making	‣ Machine decision-making processes ‣ Cybersecurity in AI
AI Project Framework	‣ Problem scoping ‣ Data acquisition ‣ Data exploration ‣ Modeling and evaluation
Python Programming	‣ Datatypes and variables ‣ Operators and conditionals ‣ Control statements and functions

2 Introduction & State of the Art of AI

History & Core AI	‣ Brief history and primary elements of AI ‣ Machine Learning (ML): definition and basic concepts ‣ Deep Learning (DL) and Neural Networks: overview and architecture ‣ Applications: image recognition, computer vision, speech recognition, information retrieval (IR)
Natural Language Processing (NLP)	‣ Overview: definition and significance of NLP in AI ‣ Key tasks: text understanding, semantic analysis, text generation, content creation ‣ Applications: machine translation, question answering / chatbots, dialogue systems (e.g., Siri, Alexa) ‣ Internet search categories: navigation, informational, transactional, investigative; voice-based search
Potential Use of AI	‣ Word processing systems and smart devices (predictive text, voice typing)

	‣ E-commerce: recommendation systems, personalized shopping, auction sites, scanner machines ‣ Social networking: AI algorithms in content delivery and user engagement
AI and Society	‣ Healthcare: improved diagnostics, personalized treatment, enhanced patient care ‣ Transportation: autonomous vehicles, intelligent transport management ‣ Disaster prediction: early warning systems, emergency response ‣ Agriculture: precision farming, crop health monitoring, yield prediction

3 Mathematics for AI

Matrices	‣ Introduction to matrices ‣ Types of matrices ‣ Operations: Addition, Subtraction, Multiplication, and Transpose
Vectors	‣ Vectors and its applications ‣ Vector arithmetic

4 Data Visualization

Data Visualization using Python	‣ Using matplotlib library ‣ *seaborn Not included*
Data Visualization using Graphs	‣ Types of Graphs ‣ Plotting on paper

Best of luck, Class XI!